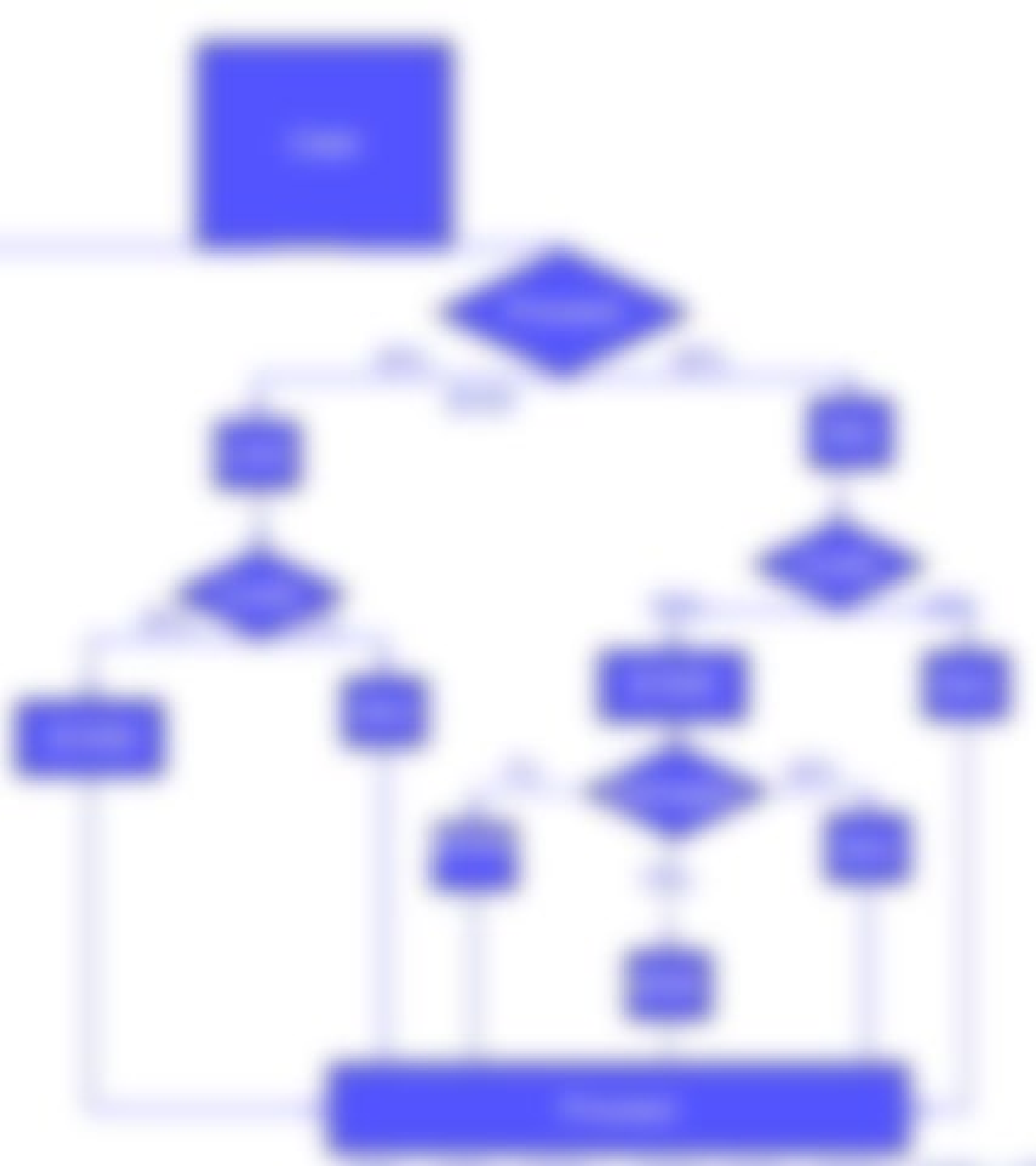


ME 300:

Engineering Thermodynamics

Example:
Phase Change
Decision Tree



Example 1

T=310 K, $v=15 \text{ m}^3/\text{kg}$ $\rightarrow$ find P , h

$\rightarrow$ Table D.1 $\rightarrow v?$ $v_g, v_f \Rightarrow v_f < v < v_g \Rightarrow$ mixture

$\rightarrow P = P_{\text{sat}} = 6.231199 \text{ kPa}$

$$\rightarrow X = \frac{v - v_f}{v_g - v_f} = \frac{15 - 0.001}{22.903 - 0.001} = 0.655$$

$$\rightarrow h = x h_g + (1-x) h_f$$

$$= (0.655)(2567.9) + (1-0.655)(154.4)$$

$$h = 1735.25 \text{ kJ/kg}$$

$$h_f < h < h_g$$

Example 2

$P=40 \text{ kPa}$, $v=0.0010264 \text{ m}^3/\text{kg}$ $\rightarrow$ find phase, T

$\hookrightarrow$ D.2

v, v_f, v_g

$\rightarrow v = v_f \rightarrow$ Phase - sat. liq.

$\rightarrow T = T_{\text{sat}} = 349.01 \text{ K}$



What phase is this fluid?

- a) Saturated vapor
- b) Saturated liquid
- c) Vapor
- d) Liquid

Example 3

$T = 286 \text{ K}$, $u = 100 \text{ kJ/kg} \rightarrow$ find phase, P , v

$\underbrace{\hspace{1.5cm}}$

D.1

$\underbrace{\hspace{3cm}}$

$u_f < u < u_g \rightarrow \text{mixture}$

$$P = P_{\text{sat}} = 1.4836 \text{ kPa}$$

$$X = \frac{u - u_f}{u_g - u_f} = \frac{100 - 53.971}{2392.6 - 53.971} = 0.0197$$

$$v = X v_g + (1 - X) v_f$$

$$= (0.0197)(88.824) + (1 - 0.0197)(0.0010006)$$

$$= 1.75 \text{ m}^3/\text{kg}$$

Example 4

$T=500\text{ K}$, $v=1.5289\text{ m}^3/\text{kg}$ $\rightarrow$ find phase, P

D.1 $v, v_f, v_g \Rightarrow v > v_g \rightarrow \text{vapor}$

$$P = 0.15\text{ MPa}$$



What phase is this fluid?

- a) Saturated vapor
- b) Saturated liquid
- c) Vapor
- d) Liquid

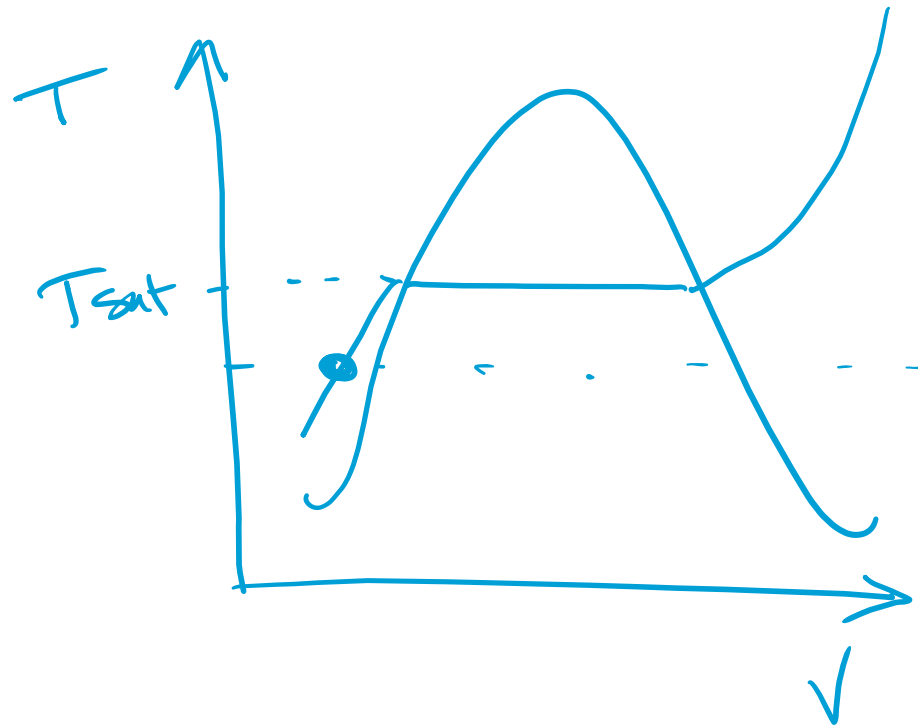
Example 5

$P=5 \text{ MPa}$, $h=117.2 \text{ kJ/kg}$ $\rightarrow$ find phase, T

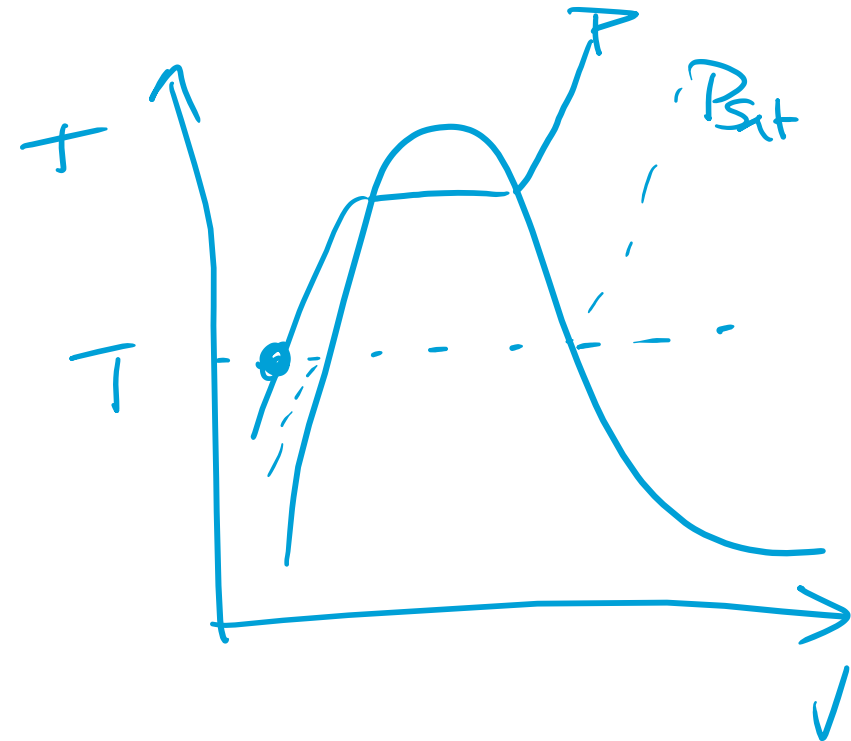
Example 6

$P=5 \text{ MPa}$, $T=420 \text{ K} \rightarrow$ find phase

$L \rightarrow D.2$ $T < T_{sat} \rightarrow \text{liquid}$



$D.1$ $P > P_{sat}$





What phase is this fluid?

- a) Saturated vapor
- b) Saturated liquid
- c) Vapor
- d) Liquid